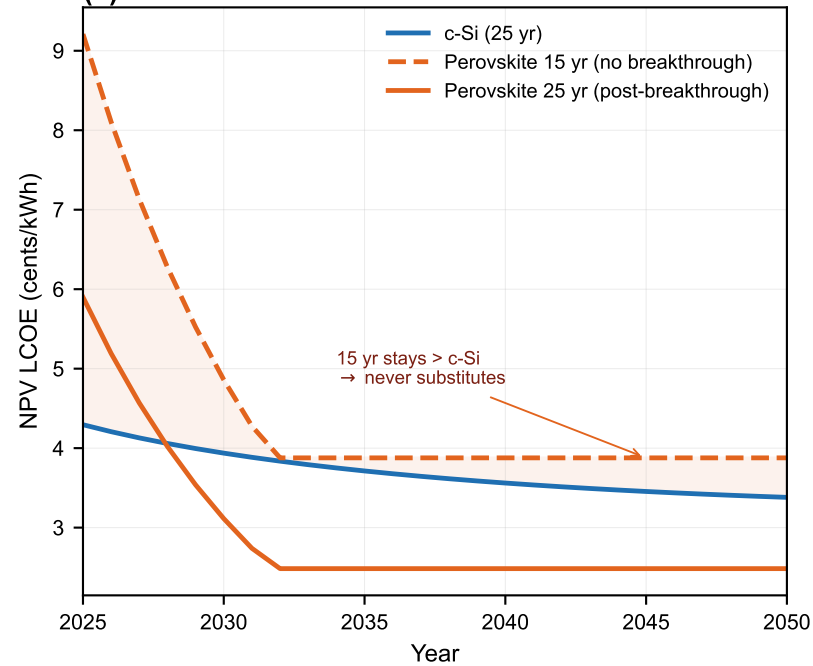
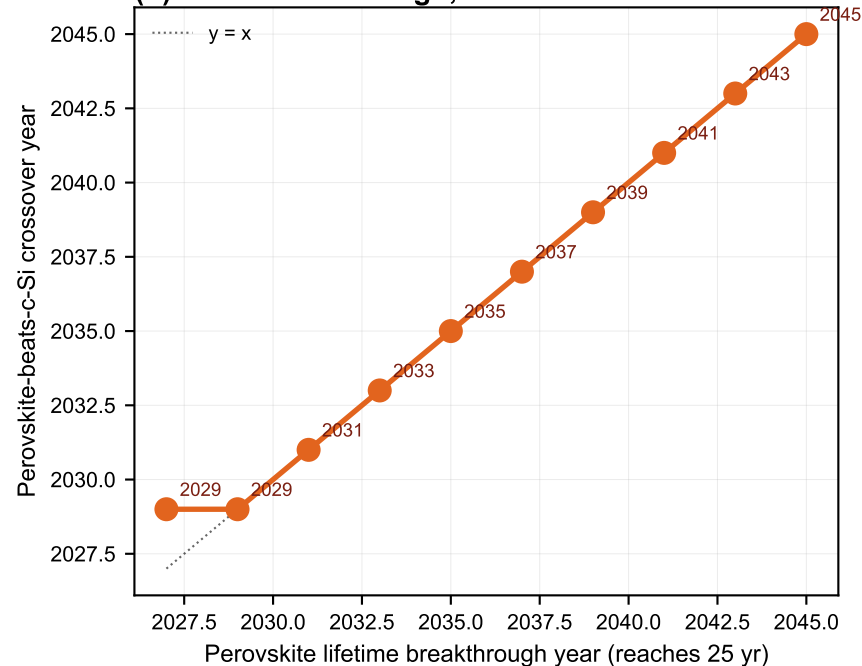


Fig 2 — Lifetime gate: what truly governs substitution (lifetime > cost)

(a) Haikou: lifetime decides



(b) Later breakthrough, later substitution



(c) Lifetime dominates cost reduction

